Contents

1 Simplicial complexes and polyhedra

Our aim is to create a good environment to prove an analogon of the Goldblatt-Thomason theorem for polyhedral semantics.

Let's begin by setting the objects we are going to find interplay between:

1.1 Simplicial complexes geometrically and combinatorially

Intuitively, any polygon can be obtained by gluing together triangles on their edges or vertices. Similarly we can describe a polyhedron as a gluing of simplices (higher dimensional triangles):

Definition 1.1 (Geometric Simplex)**.**

*The **standard n-simplex** in $\mathbb{R}^{n+1}$ is the set*

$$\Delta^n = \left\{x \in \mathbb{R}^{n+1} : x_0 + ... + x_n = 1,\ x_i \geq 0 \text{ for } i = 0, ..., n\right\}$$

endowed with the topology induced by $\mathbb{R}^{n+1}$.

Note that every standard n-simplex comes equipped with $n+1$ 'canonical' subsimplices, those given by the intersection with the canonical hyperplanes:

$$\Delta^n \geq \Delta_i^{n-1} = \left\{x \in \mathbb{R}^{n+1} : x_0 + ... + \hat{x}_i + ... + x_n = 1,\ x_i \geq 0 \text{ for } i = 0, ..., n\right\}^1$$

*We call such subsimplices and all their such subsimplices **faces** of Δ^n*

*We call an **n-simplex** in $\mathbb{R}^m$ an affine immersion $\Delta^n \hookrightarrow \mathbb{R}^m$, i.e. (the image of) an affine map*

$$j(\vec{x}) = A\vec{x} + q : \mathbb{R}^{n+1} \to \mathbb{R}^m$$

restricted to the simplex.

With this definition, we have what we expect: a 0-simplex is a point, a 1-simplex is a segment, a 2-simplex is a triangle, etc.

We can obtain a polyhedra by gluing together simplices:

Definition 1.2 (Polyhedron)**.**

*A **(Geometric) simplicial complex** is a collection K of simplices in $\mathbb{R}^m$ such that*

1. *Every face of a simplex in K is in K,*
2. *The intersection of two simplices of K is either empty or a face of both simplices.*

*We call a **Polyhedron** in $\mathbb{R}^m$ the union of all elements of a geometric simplicial complex.*

*If K is a simplicial complex we call $|K|$ its **realization**, i.e. its associated polyhedron, and given a polyhedron P, any simplicial complex K such that $|K| = P$ is called a **triangulation** of P.*

*The topology we consider is the so called **Weak** topology, i.e. the one generated by the opens of all the faces contained in it.*

Lemma 1.3. *If K is finite, then $|K|$ is compact*

Proof. It's a closed and bounded subset of $\mathbb{R}^n$. □

We make polyhedra into a category via **PL**−homeomorphisms: We consider maps that are affine when restricted to any face. More details can be found at [Col72] or (closer to the aim of the paper [Bez+21], [tGS09])

We may notice that a simplex is uniquely determined by its vertices, and a simplicial complex is uniquely determined by its simplices.

This means that we can give an abstract definition

Definition 1.4 (Simplicial Complex)**.**

*A **(Combinatorial) simplicial complex**[2] is a collection of finite sets K called **simplices**, that is closed under taking subsets, or equivalently, is a set of vertices V together with a collection $K \subseteq \mathscr{P}(V)$ (once again, of **simplices**), closed under taking subsets.*

We make simplicial complexes into a category by saying that a map of simplicial complexes is a function $V \to V'$ such that the image of any simplex is a simplex.

[1] here the hat means 'that component is removed'.

[2] The standard nomenclature is **Abstract simplicial complex**: I found that "combinatorial" fit the description better and created a more interesting duality with the geometric counterpart

Lemma 1.5. *Every geometric simplicial complex uniquely determines a combinatorial simplicial complex*

Proof. Let K be a geometric simplicial complex. Construct K^v by sending each simplex in K to the set of vertices of K.

This is a combinatorial simplicial complex, since if $\{v_1, ..., v_n\} \in K^v$ and $\{w_1, ..., w_k\} \subseteq \{v_1, ..., v_n\}$, then $\{w_1, ..., w_k\}$ are the vertices of a face of K, thus an element of K^v. $\square$

Lemma 1.6. *Every finite combinatorial simplicial complex determines a geometric simplicial complex up to PL-isomorphism.*

Proof. We can realize every simplex to the standard simplex of the correct dimension, then glue them according to the data of the simplicial complex:

$$\tilde{K} = \frac{\coprod_{\sigma \in K} \Delta^{|\sigma|}}{\sim}$$

Where $\Delta_i \sim \Delta_j$ if they correspond to the same simplex in K.

The realization of this complex only depends on the immersion of its faces, thus any two will be isomorphic, and the total map will be affine when restricted to the faces, thus PL. $\square$

From now on we refer to "combinatorial simplicial complexes" as just "simplicial complexes".

The previous lemmas mean that we have a correspondence

$$\{\text{Simplicial complexes}\} \to \{\text{Polyhedra}\}$$

We can extend it to a functor by noticing that

Lemma 1.7. *Every simplicial map $K \to K'$ induces a PL map $|K| \to |K'|$*

sketch of proof. Notice that every morphism of simplices can be realized as an affine map between their realizations. $\square$

Thus we have what is commonly known as the realization:

Definition 1.8 (Realization Functor)**.** *Call **Scplx** the category of simplicial complexes and **Pol** the category of polyhedra described above, We call the functor realizing a simplicial complex into a polyhedron*

$$|-| : \textit{\textbf{Scplx}} \to \textit{\textbf{Pol}}$$

*The **realization functor**.*

1.2 The face poset and nerves.

Notice that any simplicial complex comes equipped with a 'face' order:

Definition 1.9 (Face poset)**.**

Let Σ be a simplicial complex. Then

- $(\Sigma, \subseteq)$ *is a poset, that we call the **Face poset** of Σ and we denote $[\Sigma]^{op}$*
- $(\Sigma, \supseteq)$ *is a poset, that we call the **Coface poset of** Σ and we denote $[\Sigma]$[3]*

Note that every morphism of simplicial complexes induces a morphism of the posets -i.e. a monotone map- by taking the image of the function. It's easy to check that we have a "sort of forgetful" functor

$$[-] : \textbf{Scplx} \to \textbf{Pos}$$

As a 'forgetful' we expect there to exist a functor in the other direction that 'freely' generates a simplicial complex expressing the information of the poset; indeed

Definition 1.10 (Nerve of a poset)**.**

*Let $(P, \leq)$ be a poset. The **Nerve** $\mathscr{N}(P)$ is a simplicial complex constructed in the following way:*

- *The **vertices** are the points of P.*

[3]the reason for this notation will become clearer later.

- *The **sides** are the 1-chains $p_0 \leq p_1$ in P.*
- *The **Triangles** are the $2-$chains $p_0 \leq p_1 \leq p_2$ in P.*
- $\vdots$
- *The **n-simplices** are the $n-$chains $p_0 \leq p_1 \leq \dots \leq p_n$ in P.*
- $\vdots$

This is clearly a simplicial complex as for any $m < n$, any sub-m-chain on an $n-$chain is part of the complex. On morphisms $\mathbb{N}$ acts like the direct image:
Given a monotone map $f : (P, \leq) \to (Q, \leq)$, if $p_0 \leq \dots \leq p_k$ is a simplex in P, $f(p_0) \leq \dots \leq f(p_k)$ for monotonicity is a chain, thus a simplex in Q.

This is clearly not an inverse of $[-]$: $\mathscr{N}[\Sigma]$ generally has more points than Σ.
Let's talk of a way we can recover information from the nerve process.

1.3 Subdivisions

Definition 1.11 (Stellar moves)**.**
Given two disjoint simplicial complexes Σ and Σ' we define:
*Their **join** $\Sigma \vee \Sigma'$ to be*

$$\{\sigma \cup \sigma' : \sigma \in \Sigma, \sigma' \in \Sigma'\}$$

*The **Star** of a face σ to be*

$$\bigstar_\Sigma(\sigma) := \{\tau \in \Sigma : \tau \cup \sigma \in \Sigma\}$$

*The **link** of a face σ to be*

$$\ell_\Sigma(\sigma) := \{\tau \in \bigstar_\Sigma(\sigma) : \tau \cap \sigma = \varnothing\}$$

*Lastly, the **deletion** of a face σ as*

$$\Sigma \smallsetminus \sigma := \{\tau \in \Sigma : \sigma \notin \tau\}$$

*and the **boundary** of a face σ as*

$$\partial\sigma = \{\tau \in \Sigma : \tau \subsetneq \sigma\}$$

*A **stellar subdivision** of a complex Σ at σ is a simplicial complex*

$$(\Sigma \smallsetminus \sigma) \cup (\{x\} \vee \partial\sigma \vee \ell_\Sigma(\sigma))$$

where x is a new vertex we are introducing.
*We call a **stellar move** of a complex either a stellar subdivision or its inverse, and we say that two complexes are **stellarly equivalent** $\Sigma \sim_\star \Sigma'$ if there exist a complex Σ'' that can be obtained by Σ or Σ' via a sequence of stellar moves.*

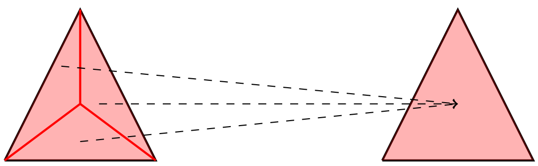
Theorem 1.12 (Alexander)**.**
*Two simplicial complexes are stellarly equivalent if and only if their geometric realizations are **PL**-homeomorphic.*

Proof. [Col72] □

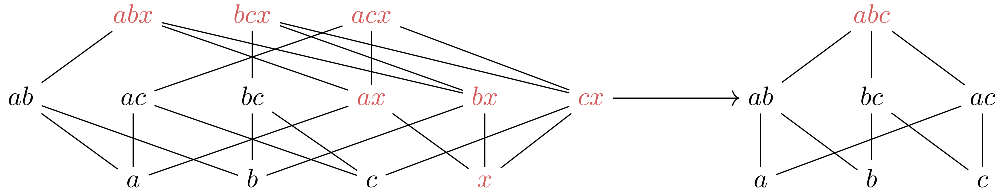
Also, a theorem that is of interest for us:

Theorem 1.13.
Stellarly equivalent simplicial complexes have bisimilar face posets.

Proof.
A complete account can be found in [Dek23]:
A brief recap is that given a simplicial complex Σ and Σ_σ a subdivision of Σ at a face σ, there exists a unique $p-$morphism $[\Sigma_\sigma] \to [\Sigma]$, i.e. the morphism sending any face $s \in \Sigma_\sigma$ to the smallest face of Σ containing s.
Here it is visually.



Or, as posets



Therefore if two simplicial complexes Σ and Σ' have a common subdivision Σ'', then there exists a p–morphism $p_1 : [\Sigma''] \to [\Sigma]$ and a p–morphism $p_2 : [\Sigma''] \to [\Sigma']$.

This means that we have a span of p–morphisms $[\Sigma] \leftarrow [\Sigma''] \to [\Sigma']$, as any span of p– morphisms defines a bisimulation, we have that $[\Sigma] \leftrightarrows [\Sigma']$. $\square$

Moreover,

Theorem 1.14 (Nerve)**.**
Let Σ be a simplicial complex. then $\mathscr{N}([\Sigma]^{op})$ is a subdivision of Σ.

Proof. This is a well known fact that can be found in [Col72], among other sources. $\square$

Later we will say $\mathscr{N}([\Sigma])$ to indicate $\mathscr{N}([\Sigma]^{\mathsf{op}})$, assuming $\mathscr{N}$ to reverse the relation whenever it's necessary: We'll see that it's more natural to talk about the coface poset, while applying the nerve yields good results when applied to the face posets.

2 Definability of compact polyhedra

[BRV01] in chapter 3 gives us a definability result for classes of finite transitive frames

Theorem 2.1. *A class of finite transitive frame is modally definable if and only if it's closed under finite disjoint union, bounded morphic images and generated subframes.*

Definition 2.2 (Modally definable class of simplicial complexes)**.**
Suppose we have a class of simplicial complexes, We say that it is modally definable if and only if the corresponding class of posets is.

We want to translate these class conditions in terms of simplicial complexes and then realize them in terms of polyhedra.

The picture we want to have in mind when building classes is the following:

$$\begin{tikzcd}
& C \arrow[dl] \arrow[d, hook] \arrow[dr] & \\
{[C]} \arrow[d, hook] & \textbf{Scplx} \arrow[dl, shift left, "{[-]}"] \arrow[dr, "|-|"] & {|C|} \arrow[d, hook] \\
\textbf{Pos} \arrow[ur, shift left, "\mathcal{N}"] & & \textbf{Pol}
\end{tikzcd}$$

Keep in mind that all kripke frames that arise in this context are posets, in particular they are transitive.

2.1 Disjoint union

Let us start with disjoint unions: these are the easiest construction since they are the same in all three settings

Proposition 2.3. *For a pair of simplicial complexes* Σ, Σ', $[\Sigma \sqcup \Sigma'] = [\Sigma] \sqcup [\Sigma']$

Proof. Let σ be a simplex of $[\Sigma \sqcup \Sigma']$, then -as the union is disjoint- it must be a simplex of either $[\Sigma]$ or $[\Sigma']$, and vice versa. □

Similarly

Proposition 2.4. *For a pair of simplicial complexes* Σ, Σ', $|\Sigma \sqcup \Sigma'| = |\Sigma| \sqcup |\Sigma'|$

Proof. There are direct ways to show it but we appeal to abstract nonsense and say that the geometric realization is the coend

$$\int^{n \in \mathbb{N}} \Delta^n \cdot \Sigma_n$$

thus the constructions commutes with colimits.

This is not as clear, but theoretically better than a direct proof in $\mathbb{R}^n$ since it eventually will allow us to work with arbitrarily-sized polyhedra too. □

This means that

Lemma 2.5. *A class of simplicial complex C is closed under disjoint union if and only if its corresponding class of posets $[C]$ is closed under disjoint union if and only if its corresponding class of polyhedra $|C|$ is closed under disjoint union.*

2.2 Generated subframes

Now suppose we have a simplicial complex Σ: What corresponds to generated subframes of $[\Sigma]$?

Recall that

Definition 2.6 (Generated subframe)**.** *A* ***Generated subframe*** *of a kripke frame (W, R), is a subset $W' \subseteq W$ closed upwards under R.*

Recall that the posets we are considering are the 'coface' posets, i.e. the posets whose elements are the faces of Σ and whose relation is the opposite of the inclusion.

This means that a generated subframe is $\uparrow$ −closed under $\supseteq$, thus $\downarrow$ −closed under $\subseteq$.

In other words

Lemma 2.7.

Let C be a class of simplicial complexes, then $[C]$ is closed under generated subframes if and only if C is closed under subcomplexes.

Proof. Let Σ be a simplicial complex and Σ' a subcomplex. As Σ' is a complex as well, then it's downward-closed.

Conversely, if $S' \leq [\Sigma]$ is a $\downarrow$-closed subposet under $\subseteq$, then by definition it's closed under taking subsets, thus it's a simplicial complex. □

Lemma 2.8. *Let C be a class of simplicial complexes, then $|C|$ is closed under subpolyhedra if and only if C is closed under subcomplexes and subdivisions.*

Proof. If $\Sigma' \leq \Sigma$ is a subcomplex then $|\Sigma'| \subseteq |\Sigma|$ is a closed subpolyhedron.

Conversely, if $S \subseteq |\Sigma|$ is a closed subpolyhedron, then there exists a triangulation Σ' of $|\Sigma|$ triangulating S, i.e. with a subcomplex $\Sigma'' \subseteq \Sigma'$ such that $|\Sigma''| = S$ (the latter is a known fact and a proof can be found in [Bez+21] as well as [Col72] or other combinatorial topology textbooks.) □

2.3 bounded morphic images

Lastly we need to interpret bounded morphism in our triad. Recall

Definition 2.9.

Let (W, R) and (W', R') be kripke frames and $f : W \to W'$ a function.
f is a ***bounded morphism*** *if and only if*

- *wRv implies $f(w)R'f(v)$*
- *$f(w)R'v'$ implies that there exists v such that wRv and $f(v) = v'$.*

Interpreting this as the coface relation this means

- $\sigma \subseteq \tau$ implies $f(\sigma) \subseteq f(\tau)$
- $\sigma' \subseteq f(\tau)$ implies that there exists $\sigma \subseteq \tau$ such that $f(\sigma) = \sigma'$

Let's define a class of simplicial maps:

Definition 2.10 (Bisimplicial map)**.**

Let $(V, \Sigma), (V', \Sigma')$ be simplicial complexes. $f : V \to V'$ is called a ***bisimplicial map*** *if*

$$\forall s \in \mathscr{P}(V), f(s) \in \Sigma' \iff s \in \Sigma$$

Lemma 2.11.

A simplicial map $f : (V, \Sigma) \to (V', \Sigma')$ induces a bounded morphism $[\Sigma] \to [\Sigma']$ if and only if it's bisimplicial.

Proof. Suppose f is a bisimplicial map, then suppose $\sigma \subseteq \tau$, then $f(\sigma)$ and $f(\tau)$ are simplices and $f(\sigma) \subseteq f(\tau)$ as it's an image map.

Now suppose $\sigma' =\subseteq f(\tau)$. Suppose $\tau = v_1, ..., v_n$, thus $f(\tau) = f(v_1), ..., f(v_n)$. Up to reordering, $\sigma' = f(v_1), ..., f(v_k)$. As f is bisimplicial, $f(v_1), ..., f(v_k) = f(v_1, ..., v_k)$ is a simplex, thus $\sigma := v_1, ..., v_k$ is a simplex and - by structure- a subsimplex of $v_1, ..., v_n$.

Now conversely, suppose that f is a bounded morphism $[\Sigma] \to [\Sigma']$, the (co)induced $f : \Sigma \to \Sigma'$ is a bisimplicial map.

First we need to check that image of a vertex is a vertex. This is because vertices are simplices, and minimal ones at that: suppose we have a vertex v such that $f(v) = \sigma'$. Then if σ' is not a vertex, there exists $w' \subsetneq \sigma' = f(v)$.

This means that there must exist $w \subseteq v$ such that $f(w) = w'$, but such w does not exist as v is minimal. $\to\leftarrow$.

Image of a simplex is a simplex as the map's domain and codomain are simplices. Suppose that $f(v_1, ..., v_n)$ is a simplex of Σ' but $v_1, ..., v_n$ is not a simplex of Σ.

applying f to $v_1, ..., v_n$ we obtain that $f(v_1, ..., v_n) = f(v_1, ..., v_n)$ (thus in particular $\subseteq$). As f is a bounded morphism defined on simplices and $f(v_1, ..., v_n)$ is an element of the image, there must exist a simplex $v_1, ..., v_k \subseteq v_1, ..., v_n$ such that $f(v_1, ..., v_k) = f(v_1, ..., v_n)$. Then, since they must have the same cardinality, it must be the case that $v_1, ..., v_k = v_1, ..., v_n$, thus $v_1, ..., v_n$ is a simplex of Σ. □

Now to interpret bisimplicial maps as maps of polyhedra:
Suppose we have the map $|f|:|\Sigma|\to|\Sigma'|$ that is the realization of a map $f:\Sigma\to\Sigma'$.
If f is bisimplicial, then this results in the conditions

- if $\mathrm{Hull}(v_1,...,v_k)\subseteq|\Sigma|$, then $f(\mathrm{Hull}(v_1,...,v_k))=\mathrm{Hull}(f(v_1),...,f(v_k))\subseteq|\Sigma'|$, and
- if $\mathrm{Hull}(f(v_1),...,f(v_k))\subseteq|\Sigma'|$, then $\mathrm{Hull}(v_1,...,v_k)\subseteq|\Sigma|$

This is equivalent to it being a retraction onto the image:

Theorem 2.12.
Suppose f if a map $\Sigma\to\Sigma'$, then it's a bisimplicial map if and only if $f:\Sigma\to\mathrm{im}(f)$ is a retraction.

Proof.
Suppose that f is bisimplicial.
First, note that f is surjective onto the vertices of $\mathrm{im}(f)$. Moreover if $\sigma'=v_1',...,v_k'$ is a simplex in $\mathrm{im}(f)$, then there exist $v_1,...,v_k$ such that $f(v_i)=v_i'$ and $v_1,...,v_k$ is a simplex in Σ.
This means that $f:\Sigma\to\mathrm{im}\,f$ is epimorphic at the level of simplices as well. Now let $g:\mathrm{im}(f)\to\Sigma$ be defined as $g(v_i')=v_i$ for some v_i such that $f(v_i)=v_i'$
This is a well-defined map, sending vertices to vertices, and $f(g(v_1',...,v_n'))=f(v_1,...,v_n)=v1',...,v_n'$ thus $fg=1_{\mathrm{im}(f)}$
Conversely, suppose that f is a section, i.e. there exists a $g:\mathrm{im}(f)\to\Sigma$ such that $fg=1_{\mathrm{im}(f)}$ Then suppose $f(v_1),...,f(v_n)=v_1',...,v_n'$ is a simplex. then $g(v_1'),...,g(v_n')=v_1,...,v_n$ is a simplex since g is simplicial. □

Corollary 2.13. *Let C be a class of simplicial complexes, then $|C|$ is closed under retracts if and only if C is closed under bisimplicial images and triangulations.*

Proof. **PL**−retracts of $|\Sigma|$ are retracts of a triangulation of Σ which are bisimplicial images by the lemma above. □

2.4 Compact Goldblatt-Thomason for polyhedra

Recall that the frames we defined are models for a modal logic (described extensively in [Bez+21]):
Essentially propositions are interpreted in unions of open subpolyhedra, $[\Diamond\phi]$ is the least closed subpolyhedron containing $[\phi]$.
We connect those as simplicial models by associating every simplex of a complex Σ to its relative interior in $|\Sigma|$, thus obtaining a neighborhood-description:

Corollary 2.14. *If Σ is a simplicial complex, for each point $x\in|\Sigma|$ there exists a unique face $\sigma\in\Sigma$ such that $x\in\sigma$, and such Σ can be interpreted as a poset through the face (or coface) relation.*

We will quickly review the fact that the semantics of polyhedra (defined in [Dek23], [Bez+21]) is equivalent to the frame semantics for the face posets of simplicial complexes modulo subdivision.
Recall

Definition 2.15 (Polyhedral semantics)**.** *Let $\mathcal{L}$ be the basic modal language. We define a polyhedral semantics as an evaluation map onto the set $\mathcal{O}_P$[4] of subpolyhedra of P open in the weak topology (defined in section 1.)*

$$\nu:\mathcal{L}\to\mathcal{O}_P$$

Thus

- $P,x\vDash\varphi\iff x\in\nu(\varphi)$,
- $\nu(\varphi\vee\psi)=\nu(\varphi)\cup\nu(\psi)$,
- $\nu(\neg\varphi)=\nu(\varphi)^c$,
- $\nu(\Diamond\varphi)=\overline{\nu(\varphi)}$.

To recollect the semantics in the literature, notice that

[4] This will annoy algebraic geometers

1. We can see $|-|$ as realizing open subpolyhedra: that is, if $\sigma \in \Sigma$ is a simplex, $|\sigma|$ is the open simplex of $|\Sigma|$ corresponding to it, this way
$$|\Sigma| = \bigsqcup_{\sigma \in \Sigma} |\sigma|$$
is a disjoint union. This means that the closure of a $|\sigma|$ is the union of $|\sigma|$ with its boundary (i.e. all the subsimplices of σ), in other words $\overline{|\sigma|} = |\downarrow \sigma|$.

2. Using the classic correspondence, we have that the coheyting algebra $\mathrm{Sub}(P)$ of closed subpolyehdra is indeed the codomain of evaluation of the cointuitionistic[5] fragment of $\mathcal{L}$
$$\{\varphi \in \mathcal{L} : \varphi \Leftrightarrow \Diamond\phi\}.$$

Let's quickly review the correspondence:

Lemma 2.16 (Polyhedral-simplicial semantic equivalence)**.**
Let Σ be a simplicial complex. Recall that for all $x \in |\Sigma|$ there exists a unique $\sigma_x \in \Sigma$ such that $x \in |\sigma_x|$. Then, employing the kripke semantics on the face poset
$$|\Sigma|, x \vDash \varphi \iff [\Sigma], \sigma_x \vDash \varphi$$

Proof. A proof can be found in [Bez+21]. □

Moreover,

Lemma 2.17.
$$\{\varphi \in \mathcal{L} : P \vDash \varphi\} = \bigcap_{\Sigma : |\Sigma| = P} \{\varphi \in \mathcal{L} : \Sigma \vDash \varphi\}$$
i.e. the logic of a polyhedron is the intersection of the logics of its triangulations.

Proof. A proof can be found in [Dek23]. □

Here we can finally express the main result of this paper:

Theorem 2.18 (Compact GTT for Polyhedra)**.**
A class of compact polyhedra is modally definable if and only if it's closed under disjoint unions, closed subpolyhedra and retractions.

Proof.
Using the previous lemmas, we know that a class of polyhedra is modally definable if and only if the class of all of its triangulations is modally definable:

Let P be an element of $\{P \in \mathbf{Pol} : P \vDash \varphi\}$, then suppose $|\Sigma| = P$, we have that for all $x \in P, P, x \vDash \varphi$, thus for all $\sigma_x \in \Sigma, \Sigma, \sigma_x \vDash \varphi$. Conversely, suppose every triangulation of P models φ, then $P \vDash \varphi$ as well, since ϕ is in the intersection of the logics of all its triangulations.

Thus, the class of polyhedra is modally definable if and only if the class of simplicial complexes formed by its triangulations (which is automatically closed under subdivision for Alexander's theorem) is modally definable, i.e. its face posets are modally definable.

The face posets are modally definable if and only if they're closed under disjoint unions, generated subframes and bounded morphic images,

this is the case if and only if the class of polyhedra is closed under disjoint unions, subcomplexes and bisimplicial images,

and this is the case if an only if the class of polyedra is closed under disjoint unions, closed subpolyhedra and retractions. □

[5] CoHeyting, paraconsistent

3 Homology

Homology is a hot topic when talking about polyhedra. Here is a quick account.

3.1 simplicial homology

We will define the homology of simplicial complexes. A full account can be found in any algebraic topology book, such as [Hat02], [Arm83].

Providing proofs of these basic facts is beyond the scope of this article, thus we will avoid them completely and refer to the forementioned books for any proof of theorems stated below.

From now on when we call a simplicial complex (V, Σ) we also fix an order on the vertices, and build simplices respecting that order.

Definition 3.1 (Simplicial k–chain)**.**
Let (V, Σ) be a simplicial complex. We call $\Sigma_k = \{\sigma \in \Sigma : \#(\sigma) = k\}$
*We call C_k the **simplicial k-chain** group*

$$\bigoplus_{\sigma \in \Sigma_k} \mathbb{Z}[\sigma],$$

i.e. the free abelian groups generated by the k–simplices of Σ.

Definition 3.2 (Boundary Map)**.**
Let (V, Σ) be a simplicial complex, $\Sigma \ni \sigma = (v_1, ..., v_k)$ be a $k - simplex$.
*We define the **boundary map***

$$\partial_k : C_k \to C_{k-1}, \ \partial_k(\sigma) = \sum_{i=1}^{k} (-1)^i (v_1, ..., \hat{v}_i, ..., v_k)$$

We define two subgroups of C_k

$$Z_k := \ker(\partial_k), B_k := \operatorname{im}(\partial_{k+1})$$

Definition 3.3 (Homology groups)**.**
*We define the **kth homology group***

$$H_k(\Sigma) = Z_k / B_k$$

Definition 3.4 (Homology chain)**.** *Together with the boundary maps, we have a long exact sequence*

$$H_0(\Sigma) \to H_1(\Sigma) \to H_2(\Sigma) \to \cdots$$

*That we call such a sequence $H_\bullet(\Sigma)$ the **homology chain** of Σ.*
We say that a morphism of homology chains $f_\bullet : H_\bullet(\Sigma) \to H_\bullet(\Sigma')$ is a collection of group homomorphisms $f_i : H_k(\Sigma) \to H_k(\Sigma')$ such that every square

$$\begin{tikzcd} H_0(\Sigma) \arrow[r] \arrow[d, "f_0"'] & H_1(\Sigma) \arrow[r] \arrow[d, "f_1"'] & H_1(\Sigma') \arrow[r] \arrow[d, "f_2"'] & \cdots \\ H_0(\Sigma') \arrow[r] & H_1(\Sigma') \arrow[r] & H_2(\Sigma') \arrow[r] & \cdots \end{tikzcd}$$

commutes.

The intuitive idea is that the rank of H_k counts *how many k–dimensional holes* our complex has, with H_0 counting connected components.

Example 3.5.

- *The empty triangle has $H_0 \simeq \mathbb{Z}$, $H_1 \simeq \mathbb{Z}$ and any subsequent $H_k \simeq 0$*
- *The filled triangle has $H_0 \simeq \mathbb{Z}$ and all other $H_k \simeq 0$.*
- *The empty tetrahedron has $H_2 \simeq \mathbb{Z}$*
- $\vdots$

3.2 Modally undefinable homology groups

While there's little hope to give much information about homology through a modal language we shall note that our constructions map neatly onto the homology chain. In particular

Proposition 3.6. $H_\bullet(\Sigma \sqcup \Sigma') \simeq H_\bullet(\Sigma) \oplus H_\bullet(\Sigma')$

Proposition 3.7. *If $\Sigma \leq \Sigma'$ is a closed subcomplex, then there's a map $H_\bullet(\Sigma) \to H_\bullet(\Sigma')$*

Proposition 3.8. *If $f : \Sigma \to \Sigma'$ is a simplicial map, then there's a bisimplicial map $f_* : H_\bullet(\Sigma) \to H_\bullet(\Sigma')$*

This means

Lemma 3.9. *The homologies of a modally definable class of polyhedra must be closed under direct sum and images.*

Or, in the more useful direction:

*If the homology groups of a class of polyhedra is **not** closed under direct sums and images, then the class is not modally definable.*

Proof. Follows from the three propositions. □

Corollary 3.10. *The class of genus n polyhedra is not modally for any positive n*

Proof. $H_1(\Sigma) \simeq \mathbb{Z}^n$ for all $\Sigma \in C$. As $\mathbb{Z}^n \oplus \mathbb{Z}^n \neq \mathbb{Z}^n$, the class is not closed under direct sums, thus not modally definable. □

4 Future Work

An interesting question is what happens when we allow for arbitrarily sized simplicial complexes, both in dimension and number of simplices. Then the class is not of transitive finite frames but arbitrary ones.

- What is the geometric realization of an arbitrarily big simplicial complex?
- What is the ultrafilter extension of an arbitrarily big simplicial complex?
- How do we describe such ultrafilter extension geometrically?
- Simplicial complexes allow for a coalgebraic description: They are a class of $\mathscr{P} \circ \mathscr{P}$ coalgebras over (finite) sets (the map $V \to \mathscr{P}\,\mathscr{P}\,V$ is the one associating to every point the up-set of simplices containing it). Do completeness and definability results coincide to the usual ones in that setting?

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